

Summary of the characteristics of influenza A H3N2, A H1N1 and B viruses isolated during 2006

Some 600 influenza viruses, isolated in 30 countries since January 2006, were characterized. These included approximately equivalent numbers of A H1N1 (31%), A H3N2 (35%) and B (34%) viruses (Table 1). A H3N2 viruses were less prevalent at the beginning of the year, but accounted for approximately half of the viruses isolated since April. The majority of the influenza B viruses were of the B/Victoria lineage (84%).

AH1N1 viruses

The majority of H1N1 viruses were antigenically closely related to the current vaccine strain A/New Caledonia/20/99 and to the more recent reference viruses A/Netherlands/128/2004 (MDCK), A/Hong Kong/2367/04 (MDCK), A/Thessaloniki/24/2005 (egg), and A/Egypt/39/2005(egg) (Table 2). A few gave reduced HI titres with the reference ferret antisera, but there was no correlation with particular amino acid changes in the HA. No consistent differences in HI were observed between HA variants of the two phylogenetic groups.

The **HA sequences** of recent isolates fell into two phylogenetic groups (Figure 1):
1) one represented by A/Thessaloniki/24/2005 and characterized by the change Y252F, many with V314A (Figure 1). There was relatively little change in the sequences of these variants over the past 12-18 months. The **NA sequences** of a number of these viruses were also shown to be similar to those of H1N1 viruses circulating during the previous year (Figure 2).
2) the other represented by A/Hong Kong/2367/2004 and A/Egypt/39/05 and characterized by the amino acid changes T82K, Y94H, R145K, R208K and T266N (Figure 1), and prominent among Asian isolates; additional changes in approx. half of the viruses include K73R plus V128T/A, many of which also have K140E (T128).

AH3N2 viruses

Many of the H3N2 viruses were antigenically closely related to MDCK-grown reference viruses A/Hong Kong/4443/2005 and A/Annecy/1138/2005 (Tables 3-5) and the more recent A/Bayern/4/2006 and A/Lyon/636/2006 (Table 6). In general, they gave low HI titres, relative to the homologous virus, with antisera to egg-grown vaccine strains, including A/Wisconsin/67/2005 and A/Hiroshima/52/2005 (Tables 3-6). These differences are due in large part to the cells in which the viruses are grown. A number of viruses gave low HI titres with all the reference antisera (e.g. Table 6). When human RBCs were used in place of turkey RBCs, HI titres were generally higher (Table 7).

HA sequences of recent isolates fell into 2 principal phylogenetic groups:

1) The majority were characterized by the changes S193F and D225N and represented by A/Hong Kong/4443/2005, A/Wisconsin/67/05 and A/Slovakia/34/2006 (Figure 3); variants were further distinguished by other prominent changes, G50E, or a combination of G78D, A198T and K309R.

2. A smaller number (approx. 20%) fell within the older phylogenetic group characterized by amino acid changes V112I and K173E and represented by A/Berlin/2/2006, A/Lyon/636/2006 or A/Bayern/4/2006; variants were distinguished by additional changes, N145S plus D188Y (or D53N), or S199P (or G50E) (Figure 3).

NA sequences also separated into 2 corresponding phylogenetic groups (Figure 4):

1) Represented by A/Wisconsin and characterized by the D93N change, relative to earlier isolates; some more recent isolates, such as A/Slovakia/34/2006, had additional changes, L370S (and R210K and P386S). 2) Represented by 2 variants, A/Bayern/4/2006, which is similar to earlier isolates, or A/Berlin/2/2006, which had the amino acid changes H150R, V194I, Y310H, L370S and S372L.

Influenza B viruses

The majority of **B/Yamagata-lineage** (HA) viruses were antigenically closely related to the recent egg-grown reference strains, B/Egypt/144/2005 and B/Florida/7/2004; many gave reduced HI titres with B/Shanghai/361/2002 and B/Jiangsu/10/2003 (Table 8).

HA and NA sequences obtained for recent isolates were mostly similar to those of viruses circulating during 2005. The majority of HAs were characterized by the change P108A, relative to the HA sequences of B/Egypt/144/2005 or A/Florida/7/2004; some have the additional change M252I (Figure 5). NA sequences of a number of these viruses had changes K125T, N340D (and T68A), relative to those of A/Egypt/144/2005 and A/Florida/7/2004 (Figure 7). HAs of a few viruses were closer in sequence to that of A/Egypt/144/2005, with additional changes N166Y and G230D, or closer to the HA of A/Jiangsu/10/2003.

Most of the **B/Victoria-lineage** (HA) viruses were antigenically closely-related to the MDCK cell-grown reference strain B/Hong Kong/45/2005 (Table 9); however, many gave reduced HI titres with the egg-grown vaccine strain B/Malaysia/2506/2005.

The majority of **HA sequences** of viruses isolated during 2006 fell into the phylogenetic group characterized by the amino acid changes K48E, K80R and K129N, relative to the sequence of B/Shandong/7/97, and represented by B/Hong Kong/45/2005 and B/Malaysia/2506/2004 (Figure 6). A small proportion fell within the other phylogenetic group characterized by the changes, K80E plus V190I, and represented by B/Ohio/1/05 (Figure 6); some have additional changes N75K and V177I, such as B/Toulouse/44/06. We did not observe any obvious differences in HI tests between viruses belonging to the different phylogenetic groups. Most of the **NA sequences** of 2006 isolates, corresponding to the former group of HAs, were close to the NA sequence of B/Ohio/1/2005 and characterized by amino acid changes E320D, L396F, E320D (and N222K), and were distinguished from the sequence of the NA of B/Malaysia/2506/2004 by 8 amino acid differences (Figure 7).

Acknowledgements

In addition to data received from the other 3 Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Denmark, Italy, Norway, Spain and Sweden. (indicated in Figures by suffixes).

Table 1: Human influenza isolates, January 2006 - present

Country	H1N1	H3N2	B	
			Yamagata lineage	Victoria lineage
January				
Algeria	1			
Czech Republic				5
Denmark			1	
Egypt	17			4
Finland		1		
France	6	2	1	4
Germany	1			
Hong Kong	5			
Iceland		2		
Ireland				1
Italy		2	1	1
Latvia		2		3
Netherlands		1	2	
Norway	4	3	4	6
Poland				1
Portugal	7			3
Russia		5		
Slovakia				2
Spain	4			3
Sweden	2	4	1	1
Switzerland	1			1
Turkey		3		
UK		1		
February				
Algeria	1			
Finland		2		2
France	5	3	3	9
Germany	4	4	1	5
Greece		1		2
Hong Kong	2	4		
Italy	18			1
Latvia	1	6	1	5
Norway	2	1	2	9
Poland				3
Portugal	22		1	15
Romania	1	3		
Russia	5	19		
Slovakia		1		
Spain	3		1	6
Sweden	2	3		1
Turkey		5	1	

Table 1: Human influenza isolates, January 2006 - present (continued)

Country	H1N1	H3N2	B	
			Yamagata lineage	Victoria lineage
March				
Algeria	1			
Austria		3		3
Czech Republic				4
Finland	1	3		2
France	2	3	1	
Germany	8	17	1	10
Hong Kong	1			2
Italy	4	1		3
Latvia	2	3		6
Norway	5	4	2	1
Poland	1	1		5
Portugal	13			3
Romania	4	1		
Russia	6	12		8
Serbia and Montenegro	1			
Spain	1	1	3	12
Sweden		1	2	1
Switzerland	1	2		1
Turkey	1	22		1
Ukraine	2	8		7
April				
Austria		1		
Croatia	1	1		
Finland		1	1	1
France	2	2	1	2
Germany		11		
Hong Kong	1			1
Italy	1			
Latvia	2	3		1
Norway	1			
Poland	1	1		
Romania		2		
Russia	3	3		
South Africa		1		
Sweden	1	3	1	1
Switzerland	1			
UK*	2	1		
May				
Hong Kong				1
Latvia	1			
Russia	1	4		1
South Africa	1	3		1
Sweden	1			1
June				
South Africa		10		1
Total =600	188	207	32	173
	31%	35%	5%	29%

Table 2. Antigenic analyses of influenza A H1N1 and H1N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹						
			Post infecton ferret sera						A/Egypt 39/05
			A/Beij	A/NC	A/Eg	A/Neth	A/Thess	A/HK	
			262/96	20/99	96/02	128/04	24/05	2637/04	
A/Beijing/262/96			1280	320	160	320	320	640	320
A/New Caledonia/20/99			160	640	320	640	640	1280	1280
A/Egypt/96/02(H1N2)			80	320	320	320	320	320	640
A/Netherlands/128/04			160	640	320	1280	1280	1280	1280
A/Thessaloniki/24/05			160	640	320	1280	1280	1280	1280
A/Hong Kong/2637/2004			40	320	160	160	160	640	640
A/Egypt/39/2005			160	640	320	640	640	1280	1280
A/Egypt/163/2006	15.1.06	E2	320	640	320	1280	1280	1280	1280
A/Stockholm/17/2006	5.2.06	.C, MDCK1 \1	160	160	160	640	320	640	1280
A/Montpellier/493/2006	6.2.06	.C, MDCK1 \1	160	640	160	1280	1280	1280	1280
A/Barcelona/261/2006	10.2.06	MDCKx \1	320	640	320	1280	1280	2560	2560
A/Hong Kong/456/2006	27.2.06	MDCK2 \1	<	80	40	160	80	160	160
A/St. Petersburg/7/2006	27.2.06	E1 \1	160	320	320	640	640	1280	1280
A/England/473/2006	Mar-06	SIAT1 \1	160	320	320	1280	640	1280	640
A/Parma/4/2006	Mar-06	MDCK1 \1	160	640	160	640	640	640	640
A/Salamanca/33/2006	Mar-06	MDCK1 \2	160	640	160	1280	1280	1280	1280
A/Algeria/G51/2006	2.3.06	MDCK0 \1	160	640	320	1280	640	1280	1280
A/Istanbul/540/2006	13.3.06	MDCKx \1	80	320	320	640	320	640	640
A/Lisbon/40/2006	14.3.06	MDCK1 \1	160	640	320	1280	1280	1280	640
A/Berlin/15/2006	15.3.06	MDCKx \1	160	640	160	1280	1280	1280	1280
A/Moscow/50/2006	17.3.06	MDCK2 \2	160	320	160	1280	640	640	640
A/Novi Sad/150/2006	17.3.06	E1 \1	80	320	320	640	320	640	1280
A/Finland/731/2006	21.3.06	MDCK2 \1	80	640	320	640	640	640	1280
A/Norway/900/2006	29.3.06	MDCK1 \1	80	640	160	640	640	640	640
A/Constanta/619/2006	31.3.06	MDCK2 \1	160	320	320	1280	640	1280	640
A/Switzerland/7425/2006	3.4.06	MDCKx \1	160	640	320	640	1280	1280	2560
A/Pol/Bydgoszcz/3/2006	5.4.06	MDCKx \1	80	160	160	160	160	320	320
A/Khabarovsk/21/2006	7.4.06	MDCK2 \2	160	320	320	1280	640	640	1280
A/Hong Kong/948/2006	10.4.06	MDCK2 \1	80	320	320	640	640	1280	1280
A/Paris/2179/2006	Apr-06	MDCKx \1	80	640	320	1280	1280	1280	1280
A/Zagreb/4622/2006	Apr-06	Ex \1	80	320	320	640	640	640	1280
A/Latvia/5172/2006	1.5.06	MDCK1 \1	80	320	320	1280	1280	1280	1280
A/Samara/128/2006	15.5.06	MDCK3 \1	160	320	160	1280	640	640	640
A/Johannesburg/131/2006	26.5.06	MDCKx \1	160	640	320	1280	1280	1280	1280

1. < : <40

Table 3. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date	Haemagglutination inhibition titre ¹									
		Post infection ferret sera									
		A/Pan	X-147	A/Shan	A/Cal	A/NY	A/Ann	A/HK	A/Anhui	A/Hiro	A/Wis
		2007/99 F7/06	(Wy/3/03) F11/03	1219/04 F14/04	7/04 F31/05	55/04 F6/05	1138/05 F33/05	4443/05 F24/05	1238/05 F1/06	52/05 F2/06	67/05 F6/06
A/Panama/2007/1999	Ex	5120	640	<	<	<	<	<	<	<	80
A/Wyoming/3/2003	Ex	2560	5120	1280	5120	1280	160	640	640	640	2560
A/Shantou/1219/2004	MDCKx	160	320	1280	2560	1280	160	640	640	320	2560
A/California/7/2004	SpfCk3, E3	40	640	2560	5120	1280	320	640	1280	320	5120
A/New York/55/2004		160	640	1280	5120	5120	160	1280	1280	640	5120
A/Annecy/1138/2005		<	160	40	640	320	320	320	320	320	640
A/Hong Kong/4443/2005		<	80	40	640	320	320	640	320	320	640
A/Anhui/1238/2005		<	80	640	640	320	160	1280	2560	1280	2560
A/Hiroshima/52/2005		80	160	320	640	320	160	1280	5120	5120	5120
A/Wisconsin/67/2005		<	320	640	640	640	320	1280	5120	5120	5120
A/Switzerland/7016/2006	21.3.06 MDCKx \1	<	<	<	160	80	80	160	320	80	640
A/Switzerland/7161/2006	24.3.06 MDCKx \1	<	80	<	320	320	80	160	160	160	640
A/Bayern/3/2006	24.2.06 MDCKx \1	<	<	<	160	80	160	160	160	80	320
A/Berlin/5/2006	6.3.06 MDCKx \1	<	40	<	640	160	320	640	640	160	640
A/Bayern/4/2006	7.3.06 MDCKx \1	<	40	<	640	320	160	320	160	80	320
A/Baden-Wuerttemberg/10/2006	28.2.06 MDCKx \1	<	<	<	160	80	80	80	80	80	160
A/Baden-Wuerttemberg/12/2006	3.3.06 MDCKx \1	<	<	<	160	80	80	160	80	160	160
A/Baden-Wuerttemberg/11/2006	2.3.06 MDCKx \1	<	40	<	320	80	160	320	160	160	320
A/Berlin/8/2006	7.3.06 MDCKx \1	<	40	<	640	80	160	320	640	80	640
A/Rheinland-Pfalz/2/2006	1.3.06 MDCKx \1	<	40	<	160	80	80	80	80	80	160
A/Berlin/11/2006	13.3.06 MDCKx \1	<	<	<	160	80	80	80	80	80	160
A/Baden-Wuerttemberg/14/2006	13.3.06 MDCKx \1	<	<	<	160	40	40	80	80	40	160
A/Berlin/12/2006	13.3.06 MDCKx \1	<	<	<	320	80	160	160	160	160	320
A/Latvia/3876/2006	24.3.06 MDCK1 \1	<	40	<	160	80	40	160	160	160	160
A/Latvia/4122/2006	29.3.06 MDCK1 \1	<	80	<	160	80	40	160	320	160	320
A/Latvia/4898/2006	23.4.06 MDCK1 \1	40	80	160	320	160	80	320	640	320	320
A/Parma/1/2006	Jan-06 MDCK1 \1	<	40	<	640	160	160	160	80	80	160
A/Parma/2/2006	Jan-06 MDCK2 \1	<	<	<	320	80	160	160	80	80	160
A/Parma/5/2006	Mar-06 MDCK1 \1	<	80	40	320	320	160	160	80	160	160
A/Lyon/CHU/3.513/2006	20.1.06 MDCK2 \2	<	<	<	40	80	<	80	80	<	160
A/Lyon/CHU/13.207/2006	28.3.06 MDCK2 \2	40	80	320	640	160	160	640	2560	320	2560
A/Lyon/320/2006	4.2.06 MDCK2 \2	<	40	<	80	40	<	80	80	80	80
A/Lyon/636/2006	27.2.06 MDCK2 \2	<	80	<	320	160	40	80	160	320	320
A/Lyon/768/2006	17.3.06 MDCK2 \1	<	80	<	160	320	160	640	1280	320	1280
A/Greece/94/2006	28.2.06 MDCK2 \1	<	<	<	160	80	40	160	160	160	320

1. <, <40

Table 4. Antigenic analyses of influenza A H3N2 viruses

			Haemagglutination inhibition titre ¹									
Viruses	Isolation		Post infection ferret sera									
	Date		A/Pan	X-147	A/Cal	A/NY	A/Ann	A/HK	A/Anhui	A/Hiro	A/Wis	
			2007/99	Wy/3/03)	7/04	55/04	1138/05	4443/05	1238/05	52/05	67/05	
			F7/06	F11/03	F31/05	F6/05	F33/05	F24/05	F1/06	F2/06	F6/06	
A/Panama/2007/1999	Ex		5120	640	<	<	<	<	<	<	<	
A/Wyoming/3/2003	Ex		1280	2560	5120	640	160	320	320	320	1280	
A/California/7/2004	SpfCk3, E3		40	1280	5120	2560	640	640	1280	320	5120	
A/New York/55/2004			160	1280	5120	5120	640	1280	1280	640	5120	
A/Annecy/1138/2005			<	80	160	160	320	160	320	320	320	
A/Hong Kong/4443/2005			40	40	160	80	80	320	320	160	320	
A/Anhui/1238/2005			40	160	1280	640	320	2560	5120	1280	5120	
A/Hiroshima/52/2005			80	320	640	640	320	2560	5120	5120	5120	
A/Wisconsin/67/2005			40	640	1280	2560	640	2560	5120	2560	5120	
A/St. Petersburg/1/2006	6.2.06	C1 \1	<	80	160	80	80	160	320	160	160	
A/St. Petersburg/2/2006	6.2.06	C1 \1	<	40	160	80	40	160	320	80	160	
A/St. Petersburg/3/2006	6.2.06	C2 \1	<	40	160	80	40	160	160	160	160	
A/St. Petersburg/4/2006	21.2.06	C1 \2	80	320	2560	2560	320	320	640	160	2560	
A/Rostov-on-Don/1/2006	23.1.06	C2 \1	40	160	640	320	160	1280	1280	640	1280	
A/Rostov-on-Don/2/2006	23.1.06	C2 \1	40	160	640	320	80	320	320	320	320	
A/Rostov-on-Don/3/2006	30.1.06	C2 \1	40	160	320	160	160	1280	1280	640	1280	
A/Kaliningrad/1/2006	30.1.06	C3 \1	<	160	640	320	80	160	320	160	320	
A/Kaliningrad/4/2006	31.1.06	C3 \2	<	80	640	640	<	80	80	40	640	
A/Kaliningrad/20/2006	28.2.06	C2 \2	<	160	640	1280	80	160	160	80	640	
A/Kaliningrad/24/2006	2.3.06	C2 \2	80	320	2560	2560	320	640	640	160	2560	
A/Astrakhan/103/2006	8.2.06	C2 \1	40	80	320	160	160	640	1280	320	640	
A/Astrakhan/150/2006	8.3.06	C3 \1	40	80	320	160	160	640	1280	320	1280	
A/Astrakhan/172/2006	15.3.06	C2 \1	<	80	640	640	40	320	160	160	320	
A/Astrakhan/319/2006	22.2.06	C2 \1	<	40	160	80	40	160	160	160	160	
A/Voronezh/1/2006	15.2.06	C2 \1	<	40	160	80	40	160	80	160	80	
A/Voronezh/4/2006	17.2.06	C2 \1	<	40	160	80	<	160	80	160	80	
A/Voronezh/6/2006	2.2.06	C2 \2	<	160	640	320	320	2560	2560	2560	2560	
A/Voronezh/10/2006	26.2.06	C2 \1	<	40	160	80	40	160	160	160	160	
A/Voronezh/18/2006	2.3.06	C2 \1	<	80	320	160	40	160	160	160	160	
A/Voronezh/58/2006	16.3.06	C2 \1	<	80	640	320	40	160	160	80	320	
A/Stavropol/2/2006	2.3.06	C2 \1	<	80	320	80	80	320	640	320	640	
A/Stavropol/3/2006	27.2.06	C2 \1	<	80	320	160	160	320	640	160	640	
A/Stavropol/5/2006	2.3.06	C2 \1	40	80	320	160	160	320	640	320	640	
A/Stavropol/6/2006	27.2.06	C2 \1	<	80	160	80	40	160	320	160	160	
A/Moscow/4/2006	19.2.06	C2 \1	<	40	160	80	<	80	80	80	80	
A/Moscow/8/2006	21.2.06	C2 \1	40	320	1280	640	80	320	320	320	640	
A/Moscow/11/2006	11.2.06	C2 \1	<	80	160	80	40	160	160	160	160	
A/Moscow/12/2006	21.2.06	C2 \1	<	80	640	320	40	160	160	80	320	
A/Moscow/16/2006	26.2.06	C2 \1	80	320	2560	320	640	2560	2560	1280	2560	
A/Finland/641/2006	14.2.06	MDCK2 \1	<	80	320	160	40	160	160	80	160	
A/Finland/682/2006	28.2.06	MDCK1 \1	<	80	640	320	40	160	160	160	160	
A/Finland/703/2006	8.3.06	MDCK1 \3	40	80	640	320	40	160	160	160	160	
A/Finland/732/2006	21.3.06	MDCK2 \1	40	80	320	160	80	320	320	320	320	
A/Finland/744/2006	29.3.06	MDCK2 \1	<	80	320	160	40	160	160	80	160	
A/Finland/772/2006	7.4.06	MDCK2 \1	<	80	320	160	40	160	160	80	160	

1. <, <40

Table 5. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date	Haemagglutination inhibition titre ¹							
		Post infection ferret sera							
		A/Pan 2007/99 F7/06	X-147 Wy/3/03 F11/03	A/Cal 7/04 F31/05	A/NY 55/04 F6/05	A/Ann 1138/05 F33/05	A/HK 4443/05 F24/05	A/Hiro 52/05 F2/06	A/Wis 67/05 F6/06
A/Panama/2007/1999	Ex	5120	640	<	<	<	<	<	<
A/Wyoming/3/2003	Ex	1280	5120	5120	640	160	320	320	1280
A/California/7/2004	SpfCk3, E3	40	640	5120	1280	320	640	320	1280
A/New York/55/2004		40	1280	5120	5120	320	1280	640	2560
A/Annecy/1138/2005		<	80	320	320	320	640	320	640
A/Hong Kong/4443/2005		<	80	320	160	80	640	640	640
A/Hiroshima/52/2005		80	320	640	320	160	1280	5120	5120
A/Wisconsin/67/2005		40	640	1280	1280	320	2560	2560	5120
A/Berlin/38/2006	? MDCKx \3	<	40	160	160	40	320	160	640
A/Berlin/37/2006	11.4.06 MDCKx \1	<	40	160	160	80	160	160	160
A/Baden-Wuerttemberg/44/2006	22.3.06 MDCKx \1	<	40	320	160	160	320	320	320
A/Berlin/23/2006	25.4.06 MDCKx \3	<	40	160	80	40	160	80	160
A/Bayern/23/2006	20.4.06 MDCKx \3	<	40	160	80	<	160	160	160
A/Baden-Wuerttemberg/35/2006	28.3.06 MDCKx \1	<	40	320	320	80	320	160	640
A/Baden-Wuerttemberg/34/2006	27.3.06 MDCKx \1	<	320	1280	640	320	2560	640	1280
A/Baden-Wuerttemberg/31/2006	22.3.06 MDCKx \3	<	80	160	80	40	160	160	160
A/Bayern/21/2006	20.3.06 MDCKx \1	<	40	320	160	80	160	160	160
A/Bayern/20/2006	16.3.06 MDCKx \3	<	40	160	80	40	160	80	160
A/Bayern/19/2006	13.3.06 MDCKx \1	<	<	320	80	80	160	160	160
A/Baden-Wuerttemberg/28/2006	10.2.06 MDCKx \1	<	40	320	160	80	160	160	160
A/Hamburg/1/2006	6.4.06 MDCKx \1	<	40	160	80	80	160	160	160
A/Berlin/32/2006	11.4.06 MDCKx \1	<	40	160	80	80	160	160	160
A/Bayern/18/2006	12.4.06 MDCKx \1	<	<	320	160	80	320	160	320
A/Thuringen/13/2006	25.3.06 MDCKx \1	<	40	320	160	80	160	160	160
A/Brandenburg/4/2006	10.4.06 MDCKx \1	<	40	160	80	80	160	160	160
A/Mecklenburg-Vorpo./4/2006	5.4.06 MDCKx \3	<	<	80	80	<	80	80	80
A/Mecklenburg-Vorpo./3/2006	4.4.06 MDCKx \1	<	40	160	80	80	160	160	160
A/Berlin/36/2006	12.4.06 MDCKx \1	<	<	160	80	80	320	160	320
A/Pol/Waw/2/2006	31.3.06 MDCKx \1	<	40	160	80	<	160	160	160
A/Pol/Bydgoszcz/1/2006	3.4.06 MDCKx \1	<	<	80	40	<	80	80	80
A/Latvia/5173/2006	28.4.06 MDCK1 \3	<	160	1280	640	40	320	80	160
A/Latvia/5653/2006	11.5.06 MDCK1 \3	<	<	80	80	<	320	80	80

1. <, <40

Table 6. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date	Haemagglutination inhibition titre ¹									
		Post infection ferret sera									
		X-147	A/Cal	A/NY	A/HK	A/Hiro	A/Wis	A/Slov	A/Berlin	A/Bay/	A/Lyon
		(Wy/3/03)	7/04	55/04	4443/05	52/05	67/05	134/06	2/06	4/06	636/06
		F11/03	F31/05	F6/05	F24/05	F2/06	F6/06	F9/06	F8/06	F11/06	F8/06
A/Wyoming/3/2003	Ex	5120	5120	640	320	320	640	640	320	1280	1280
A/California/7/2004	SpfCk3, E3	1280	5120	1280	640	320	1280	320	1280	5120	2560
A/New York/55/2004		640	5120	5120	640	640	1280	640	1280	5120	2560
A/Hong Kong/4443/2005		80	320	160	640	640	640	320	80	320	80
A/Hiroshima/52/2005		320	640	320	1280	5120	5120	320	320	2560	640
A/Wisconsin/67/2005		640	1280	1280	2560	2560	5120	640	320	2560	1280
A/Slovakia/134/2006		80	320	160	320	320	640	160	80	320	320
A/Berlin/2/2006		80	80	40	80	80	80	40	40	40	40
A/Bayern/4/2006		80	640	320	320	320	320	160	160	320	160
A/Lyon/636/2006		160	320	320	320	320	320	160	160	320	320
A/Samara/120/2006	3.5.06 MDCK3 \2	160	1280	320	160	320	160	80	160	320	320
A/Samara/123/2006	15.5.06 MDCK4 \2	160	640	320	160	320	320	160	40	320	160
A/Murmansk/3/2006	28.4.06 MDCK3 \3	160	640	320	160	320	160	160	320	320	160
A/Moscow/41/2006	16.3.06 MDCK2 \2	80	320	160	320	320	320	40	<	160	160
A/Moscow/40/2006	16.3.06 MDCK2 \3	80	640	160	160	160	160	40	80	320	320
A/N. Novgorod/155/2006	7.4.06 MDCK3 \2	80	320	320	160	320	160	80	80	640	320
A/Voronezh/19/2006	10.3.06 MDCK2 \2	160	1280	640	320	320	320	160	320	640	320
A/Voronezh/12/2006	1.3.06 MDCK2 \2	160	640	320	640	640	640	320	80	320	160
A/St. Petersburg/37/2006	8.5.06 MDCK2 \1	80	320	160	160	320	320	<	<	80	<
A/Kaliningrad/44/2006	6.4.06 MDCK2 \3	80	640	320	320	320	160	40	40	320	160
A/Khabarovsk/49/2006	22.5.06 MDCK2 \2	160	1280	640	320	640	640	160	320	640	320
A/Khabarovsk/10/2006	7.4.06 MDCK5 \2	80	640	320	160	160	160	40	80	320	320
A/Khabarovsk/29/2006	20.4.06 MDCK3 \3	80	320	160	160	320	160	80	80	320	160
A/Ryazan/10/2006	6.4.06 MDCK2 \3	160	640	320	160	160	160	80	160	640	320
A/Ryazan/4/2006	6.4.06 MDCK2 \2	160	1280	640	640	320	320	160	320	1280	320
A/Odessa/52/2006	23.3.06	40	160	160	160	80	160	40	80	320	80
A/Odessa/62/2006	27.3.06	80	320	160	160	160	160	80	80	320	80
A/Zagreb/4274/2006	Apr-06 Ex \2	40	160	80	320	160	640	80	40	640	320
A/Paris/1889/2006	Mar-06 MDCKx \1	<	160	80	320	160	160	40	<	<	<
A/Paris/2030/2006	Apr-06 MDCKx \1	<	160	80	320	160	160	40	<	<	<
A/Paris/2109/2006	Apr-06 MDCKx \1	<	40	40	160	160	80	<	<	<	<
A/Paris/2208/2006	Feb-06 MDCKx \1	80	640	640	320	160	160	80	160	640	320
A/Johannesburg/350/2006	9.6.06 MDCKx \3	<	80	80	160	160	160	<	<	<	<
A/Johannesburg/354/2006	8.6.06 MDCKx \3	<	80	80	160	160	160	<	<	<	<
A/Johannesburg/185/2006	29.5.06 MDCKx \3	<	80	80	160	160	160	<	<	<	<
A/Johannesburg/245/2006	2.6.06 MDCKx \1	40	80	80	160	160	160	40	<	<	<
A/Johannesburg/271/2006	4.6.06 MDCKx \2	40	80	80	160	80	80	40	<	80	40
A/Johannesburg/277/2006	5.6.06 MDCKx \1	<	80	80	160	160	160	40	<	80	<
A/Johannesburg/339/2006	7.6.06 MDCKx \1	<	40	40	80	40	40	40	<	<	<
A/Johannesburg/340/2006	Jun-06 MDCKx \2	<	80	40	80	40	80	<	<	<	<
A/Johannesburg/344/2006	8.6.06 MDCKx \2	<	80	40	80	80	80	40	<	<	<
A/Hong Kong/201/2006	2.2.06 MDCK2 \2	<	40	40	80	80	80	40	<	<	<
A/Hong Kong/453/2006	22.2.06 MDCK2 \2	<	80	80	80	80	80	40	<	80	<

1. <, <40

Table 7. Antigenic analyses of influenza A H3N2 viruses (Human red blood cells)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			A/Cal	A/HK	A/Wis	A/Slov	A/Berlin	A/Bay/	A/Lyon
			7/04 F31/05	4443/05 F24/05	67/05 F6/06	134/06 F9/06	2/06 F8/06	4/06 F11/06	636/06 F8/06
A/California/7/2004		SpfCk3, E3	5120	640	640	320	640	2560	1280
A/Hong Kong/4443/2005			640	1280	1280	320	40	640	320
A/Wisconsin/67/2005			320	1280	2560	320	40	320	320
A/Slovakia/134/2006			640	1280	1280	320	80	640	640
A/Berlin/2/2006			320	160	80	40	40	160	40
A/Beryern/4/2006			640	160	160	40	320	640	320
A/Lyon/636/2006			640	320	320	80	320	640	320
A/Stokholm/13/2006	04.4.06	0 \1	160	320	320	80	<	160	40
A/Stokholm/10/2006	13.2.06	0 \1	1280	160	160	80	320	1280	320
A/Paris/2208/2006	Feb-06	MDCKx \1	640	160	80	80	320	640	160
A/Umea/1/2006	23.1.06	1 \1	640	80	80	40	160	320	160
A/Johannesburg/277/2006	5.6.06	MDCKx \1	320	640	320	160	40	160	80
A/Johannesburg/350/2006	9.6.06	MDCKx \3	320	640	640	320	40	320	160
A/Paris/1889/2006	Mar-06	MDCKx \1	160	640	640	160	40	160	80
A/Paris/2030/2006	Apr-06	MDCKx \1	320	640	640	320	40	320	80
A/Johannesburg/354/2006	8.6.06	MDCKx \3	320	640	640	320	40	320	160
A/Warsaw/2/2006	31.3.06	MDCKx \1	320	320	160	80	40	80	40
A/Johannesburg/340/2006	Jun-06	MDCKx \2	320	320	320	160	<	160	80
A/Hong Kong/453/2006	22.2.06	MDCK2 \2	320	320	320	160	40	160	80
A/Johannesburg/344/2006	8.6.06	MDCKx \2	320	640	640	320	40	320	80
A/Odessa/52/2006	23.3.06	MDCKx \1	640	160	160	80	160	640	160
A/Calarasi/673/2006	6.4.06	MDCK2 \1	320	320	640	40	<	160	80
A/Bucharest/500/2006	14.3.06	MDCK2 \1	1280	1280	2560	320	160	2560	1280
A/Moscow/41/2006	16.3.06	MDCK2 \2	320	320	320	160	40	160	80
A/Samara/123/2006	15.5.06	MDCK4 \2	640	160	160	80	160	320	160
A/St.Petersburg/37/2006	8.5.06	MDCK2 \1	80	160	160	80	<	40	40
A/Hong Kong/20/2006	2.2.06	MDCK2 \2	160	320	320	80	<	80	40
A/Stockholm/16/2006	21.4.06	1 \1	640	640	640	160	40	640	320
A/Kaliningrad/24/2006	2.3.06	C2 \2	640	80	80	40	160	640	160
A/Voronezh/12/2006	1.3.06	MDCK2 \2	160	320	320	160	40	80	80
A/Paris/2109/2006	Apr-06	MDCKx \1	320	320	320	80	40	160	80
A/Umea/3/2006	31.1.06	1 \1	640	160	160	80	160	320	160
A/Stockholm/8/2006	07.2.06	0 \1	640	1280	640	320	<	320	160
A/Umea/2/2006	30.1.06	1 \1	320	80	80	80	160	320	160

1. <, <40

Table 8. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Isolation date		Haemagglutination inhibition titre				
			Post infection ferret sera				
			B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03	B/Eg 144/05	B/FI 7/05
B/Sichuan/379/99			160	320	40	160	160
B/Shanghai/361/2002			160	640	80	160	320
B/Jiangsu/10/2003			40	40	320	80	80
B/Egypt/144/2005			160	320	80	320	320
B/Florida/7/2005			80	320	80	160	320
B/Kiev/103/2005	Jan 06	MDCKx \1	160	320	320	80	320
B/Denmark/1/2006	9.1.06	MDCK2 \1	40	40	40	320	160
B/Genoa/1/2006	23.1.06	MDCK1 \2	80	80	40	160	160
B/Parma/1/2006	Feb-06	MDCK1 \1	40	160	160	160	160
B/Poitiers/746/2006	Feb-06	, MDCK1 \1	40	160	320	320	160
B/Bursa/197/2006	1.2.06	MDCKx \1	40	160	160	160	320
B/Lisbon/23/2006	4.2.06	MDCK2 \1	160	320	160	640	640
B/Niedersachsen/14/2006	27.2.06	MDCKx \1	40	320	160	160	320
B/Soria/44/2006	Mar-06	MDCKx \1	40	160	80	160	320
B/Bucharest/504/2006	14.3.06	MDCK2 \1	80	160	80	80	160
B/Norway/704/2006	20.3.06	MDCK1 \1	160	320	160	320	640
B/Paris/2082/2006	Apr-06	MDCKx \1	40	80	80	160	160
B/Finland/767/2006	8.4.06	MDCK2 \1	80	160	160	160	320
B/Stockholm/8/06	19. 4.06	MDCK1 \1	160	160	80	160	160

Table 9. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Isolation date		Haemagglutination inhibition titre			
			B/Shan ² 7/97	Post infection ferret s		
				B/Shan 7/97	B/Bris 32/02	B/HK 45/05
B/Shandong/7/97			1280	320	320	160
B/Brisbane/32/2002			1280	160	320	160
B/HK/45/2005			1280	320	320	160
B/Malaysia/2506/2004			1280	160	320	160
B/Ireland/2193/2006	Jan-06	MDCKx \1	1280	160	40	160
B/Netherlands/002/2006	19.1.06	MDCK1 \1	2560	320	40	160
B/Czech/508/2006	20.1.06	TC1, E1 \1	1280	160	320	320
B/Egypt/205/2006	22.1.06	C1 E1	1280	160	40	80
B/Norway/196/2006	24.1.06	MDCK1 \1	1280	160	40	160
B/Slovakia/117/2006	30.1.06	MDCK1 \1	1280	40	40	160
B/Genoa/2/06	30.1.06	MDCK1 \1	1280	160	<	320
B/Lisbon/13/2006	5.2.06	MDCK1 \1	1280	80	40	320
B/Lyon/422/2006	8.2.06	MDCK2 \1	1280	160	<	80
B/Barcelona/258/2006	10.2.06	MDCKx \1	1280	80	<	160
B/Greece/68/2006	21.2.06	MDCK1 \1	1280	160	80	80
B/Napoli(ISS)/4/2006	Mar-06	MDCK1 \1	1280	80	<	160
B/Finland/675/2006	2.3.06	MDCK1 \1	1280	80	40	160
B/Pol/Waw/8/2006	2.3.06	MDCKx \1	640	160	<	80
B/Switzerland/6299/2006	3.3.06	MDCKx \1	1280	320	320	160
B/Sachsen/2/06	7.3.06	MDCKx \1	1280	80	40	160
B/Istanbul/506/2006	8.3.06	MDCKx \1	1280	160	40	80
B/Kiev/366/2006	18.3.06	MDCK \1	1280	160	40	80
B/St. Petersburg/17/2006	20.3.06	C2 \2	640	160	80	80
B/Austria/281388/2006	23.3.06	MDCKx \1	640	160	<	160
B/Paris/2146/2006	Apr-06	MDCKx \1	1280	160	<	160
B/Latvia/4255/2006	4.4.06	MDCK1 \1	1280	80	40	160
B/Valladolid/56/2006	6.4.06	MDCKx \1	640	80	<	80
B/Stockholm/7/06	11.4.06	MDCK1 \1	1280	80	160	160
B/Samara/99/2006	26.4.06	MDCK4 \3	640	160	160	80
B/Ryazan/4/2006	11.5.06	MDCK2 \1	1280	80	80	160
B/Hong Kong/809/2006	11.5.06	MDCK2 \1	640	160	<	160
B/Johannesburg/282/2006	1.6.06	MDCKx \2	640	160	<	160

¹ <, <40

² hyperimmune sheep serum

Figure 1. Phylogenetic comparison of H1 HAs

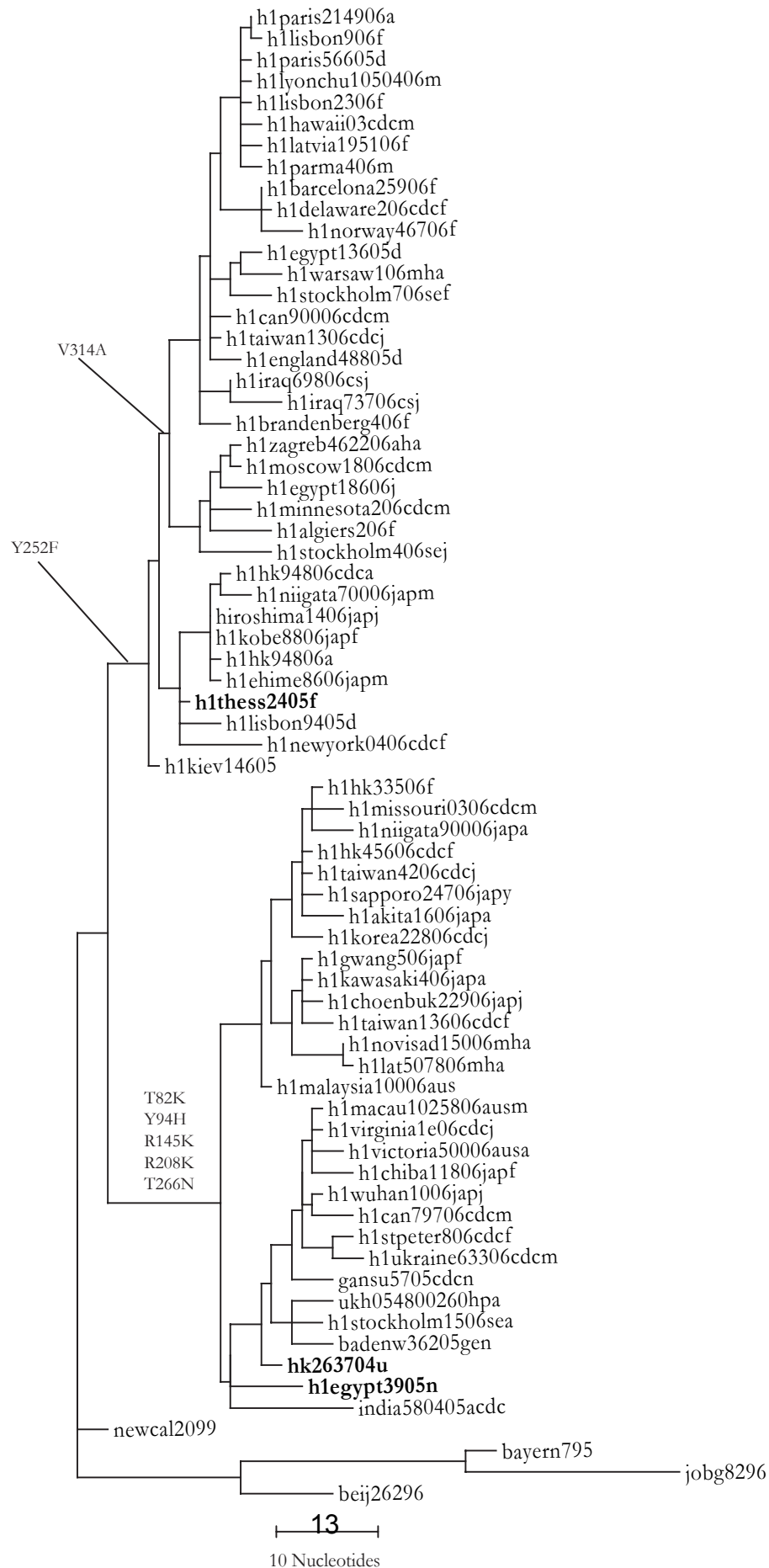


Figure 2. Phylogenetic comparison of N1s

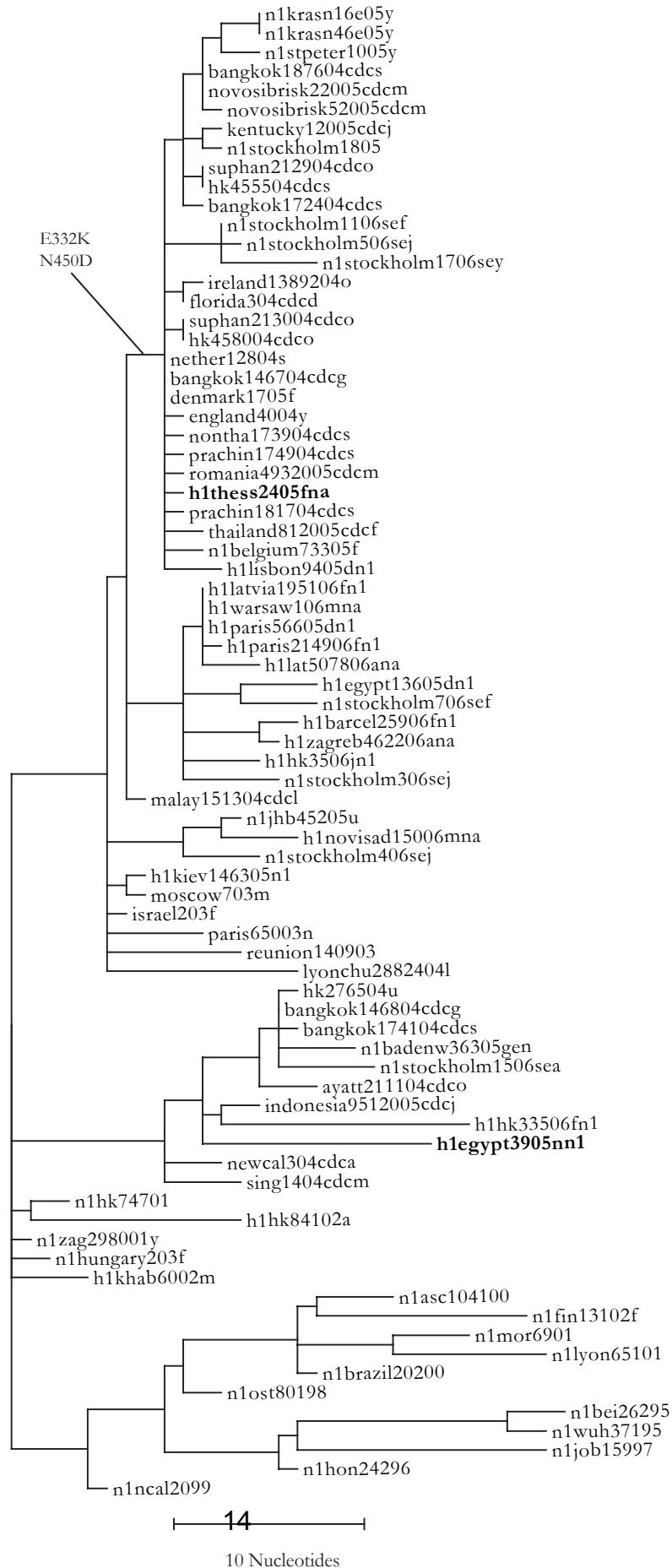


Figure 3. Phylogenetic comparison of H3 HAs

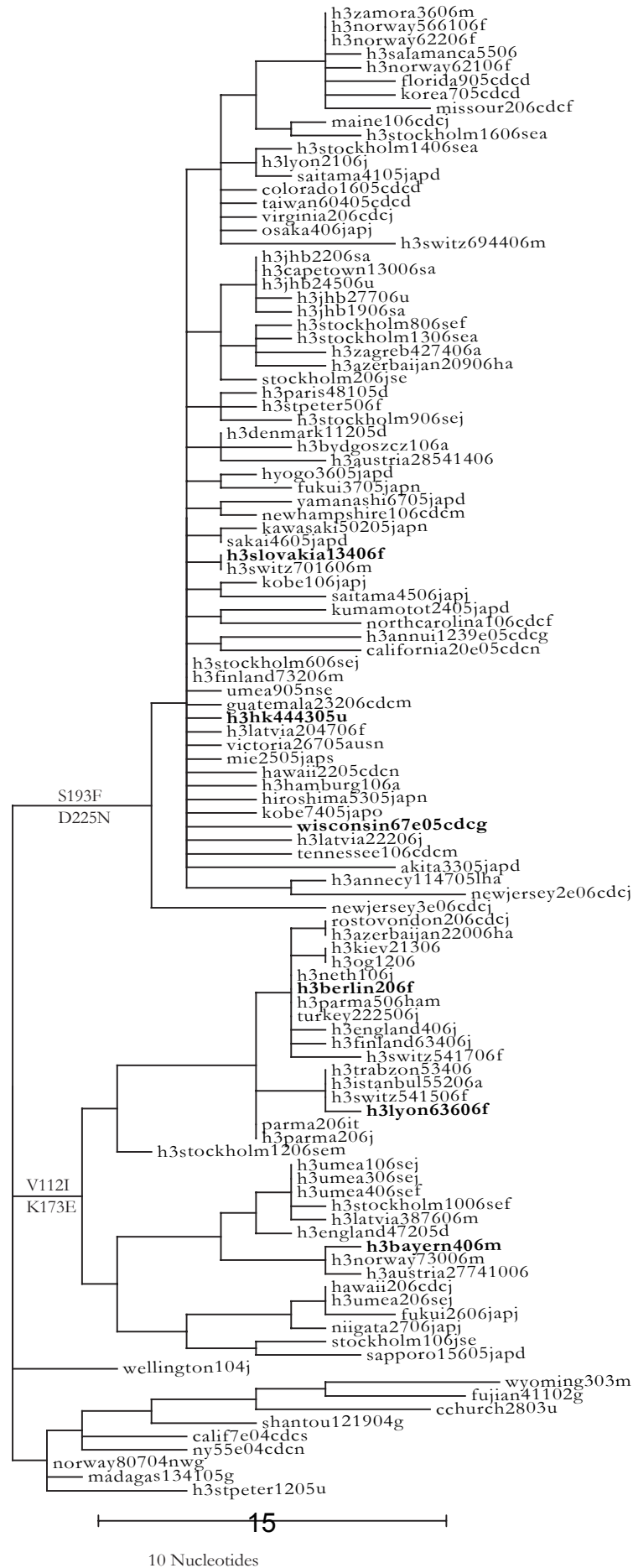


Figure 4. Phylogenetic comparison of N2 NAs

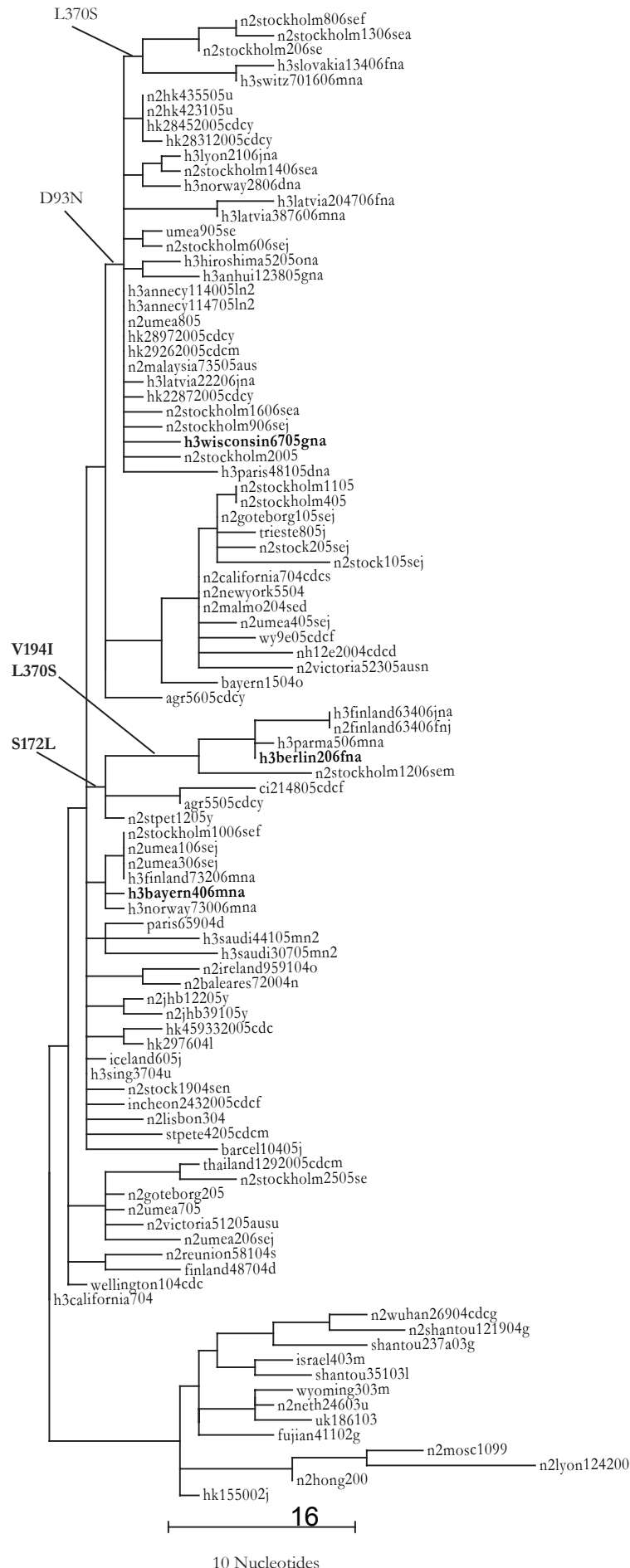


Figure 5. Phylogenetic comparison of BHAs (Yamagata lineage)

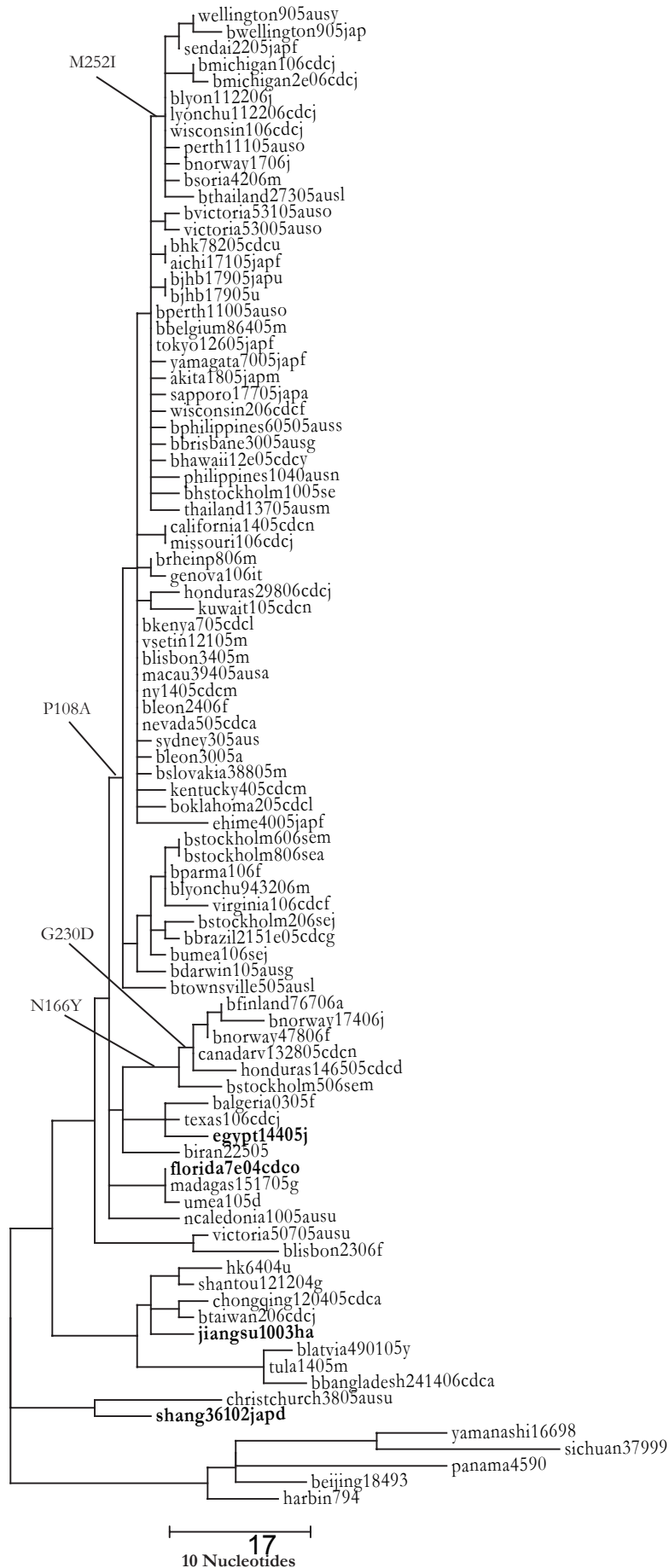


Figure 6. Phylogenetic comparison of BHAs (Victoria lineage)

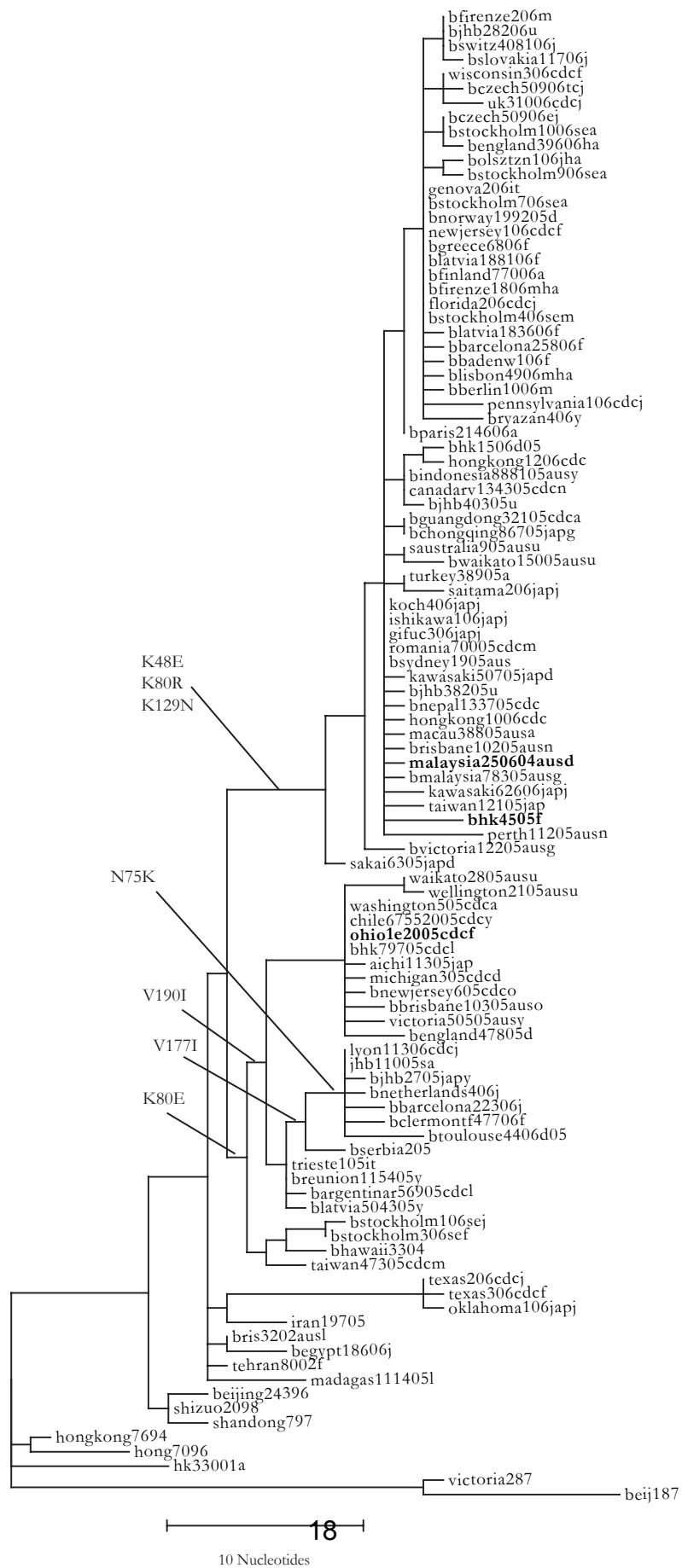


Figure 7. Phylogenetic comparison of BNAs

